

Classification Documentation: Drug Consumption Dataset.

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1. Classification Task and Feature Selection

Building on the *Exploratory Data Analysis*, the central question this phase targets is **whether personality traits and demographic characteristics can predict cannabis consumption**.

The target variable was defined by *binarizing the Cannabis column at the CL0 boundary*: composing the **non-user class** with respondents that reported CL0, while CL1 through CL6 compose the **user class**. Binarization, rather than direct use of the seven-level CL scale, is necessary because Cannabis is measured on an ordinal scale, where arithmetic on the labels has no meaning. The resulting split is approximately **76% Users and 24% Non-users**, a moderate imbalance that motivates the use of the weighted F1-score as the primary evaluation metric.

The independent variables are composed of the *seven NEO-PI-R personality scores* and the *five demographic features*. All drug columns were excluded as predictors: including them would constitute data leakage, since a model predicting cannabis from cocaine use would learn the trivial co-occurrence of substance use rather than the personality and demographic link that the research question targets. Cannabis itself was excluded because it is used to derive the target.

Variables were encoded according to their measurement scale: *personality scores* were kept as raw z-scores, *Age* and *Education* were mapped to integers preserving the natural order from youngest to oldest and from least to most education. *Gender*, *Country* and *Ethnicity*, being nominal, were one-hot encoded.

For the **k-Nearest Neighbor classifier**, all features were additionally rescaled via Min-Max normalization. Without this, the personality scores would dominate over the binary dummies. The **Decision Tree** did not require scaling because splits are invariant to monotonic transformations of individual features.

2. Methodology

Two classifiers from distinct paradigms were compared: a **Decision Tree**, which is *rule-based*, interpretable and outputs native feature importance, and a **k-Nearest Neighbor** classifier, which is *instance-based* and *distance-driven*. This contrast produces feature-importance rankings from two methodologically independent perspectives and provides a rich analysis over the dataset.

Both classifiers were evaluated using **stratified 5-fold cross-validation**: the data is partitioned into five folds preserving the 76/24 class ratio, each fold is used once for testing while the other four train the model, and the resulting five scores are averaged. The standard deviation of these five scores is reported alongside the mean and is the more important number of the two: it indicates how stable the model is across different splits of the data, as the standard deviation of the average is more reliable than the score of a fold affected by an unlucky split. The **weighted F1-score** was used as the selection metric for hyperparameter tuning, as accuracy is misleading under class imbalance.

For the **Decision Tree**, the split criterion was set to entropy, and the maximum depth was optimized over a range from 2 to 15 by maximizing the mean cross-validated F1-score. The optimum depth was obtained at 8, with a mean weighted F1 of 0.770. For the **k-Nearest Neighbor** classifier, the number of neighbors was tuned over odd values from 1 to 51 to avoid tied votes. The optimum neighbor number was obtained at 7, with a mean weighted F1 of 0.794.

3. Results

The two classifiers were evaluated using accuracy, the weighted F1-score, and class-conditional precision and recall. Table 1 summarizes the results. The **kNN classifier outperforms the Decision Tree** on both summary metrics. A trivial baseline that always predicts the majority class would achieve 78.1% accuracy,

the Decision Tree at 77.1% sits within the standard deviation of this baseline, while kNN at 80.4% presents a real improvement over the baseline. The low standard deviations indicate that **both results are stable**.

Classifier	Accuracy	Weighted F1
Decision Tree (depth = 8)	0.7712 ± 0.0155	0.7697 ± 0.0133
kNN (k = 7)	0.8041 ± 0.0123	0.7935 ± 0.0143

Table 1: Five-fold cross-validation performance (mean $\pm$ standard deviation).

The asymmetry between the two classes is more revealing than the headline scores. Figure 1 makes this visible: more than half of the actual Non-users are misclassified as Users by both models. In practical terms, the classifiers reliably identify Cannabis users, when either model predicts User, that prediction is correct around 85% of the time, and around 86% (Tree) to 91% (kNN) of actual users are caught, but they struggle with non-users, achieving recall of only 0.47 (Tree) and 0.43 (kNN). A prediction of Non-user is therefore much less trustworthy than a prediction of User.

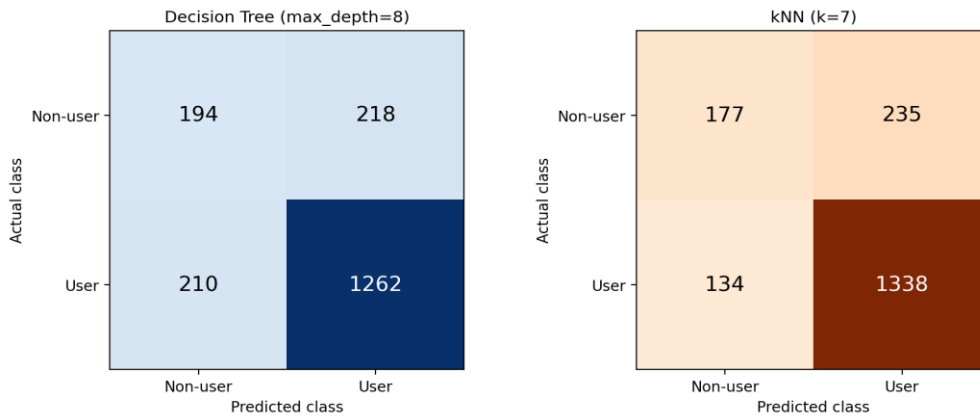


Figure 1: Out-of-fold confusion matrices for the two classifiers (5-fold CV).

4. Feature Importance

Feature importance was computed using two complementary methods. The Decision Tree was analyzed via its native **impurity-based importance**, which accumulates the entropy reduction attributable to each feature. The kNN classifier was analyzed via **permutation importance**, the only feasible method for an instance-based model without internal feature weights. Table 2 reports the top three features under each ranking.

Rank	Decision Tree (impurity)	kNN (permutation)
1	Country_UK	Sensation Seeking (SS)
2	Sensation Seeking (SS)	Age
3	Conscientiousness (Cscore)	Openness (Oscore)

Table 2: Top three features by importance for each classifier.

The two rankings agree on Sensation Seeking as a top predictor, which is also the trait flagged most prominently by the exploratory analysis. Beyond that, the rankings diverge:

kNN places Age and Openness in second and third positions, which directly confirms the EDA hypothesis that younger respondents with elevated Openness are more likely to be Cannabis users.

The Decision Tree, by contrast, places Country_UK in first position. This is possibly a shortcut rather than a causal signal: UK respondents account for 55% of the sample, so the dummy variable gives it priority.

The convergence on Sensation Seeking across two methodologically independent models is therefore the most reliable indicator of true predictive signal.

5. Conclusions

The classification confirms that personality and demographic features contain genuine predictive signal for Cannabis use, but it also reveals the structural limitations of this dataset. To test these limits, two hypothetical respondents were constructed: **Respondent A**, a young, university-educated female from the UK with high Sensation Seeking and Openness, and **Respondent B**, an older male from the USA with a doctorate and low Sensation Seeking and Openness.

Both classifiers **correctly predict Respondent A as a User** with high confidence. Both also **predict Respondent B as a User**, contrary to the EDA expectation.

A diagnostic analysis of the seven nearest neighbors of Respondent B in the scaled feature space explains this result: even among the most similar respondents, the 76% correspondent to the User class is highly dominant. The differences in personality scores conditioning is real but modest, typically between 0.3 and 0.7 standard deviations, which is insufficient to affect the predictions in borderline cases.

The conclusion is not that the models are broken but that the dataset does not support this level of analysis for the chosen features. The models reliably identify Cannabis users but identify Non-users less than half the time.

Three findings stand out:

1. The EDA hypothesis that Sensation Seeking, Openness and Age are linked to Cannabis use is supported by both classifiers and remains the most reliable signal across importance methods.
2. The class imbalance and moderate effect sizes produce a bias that cannot be overcome by classifier choice alone.
3. The divergent feature rankings between the Decision Tree and kNN illustrate that importance is model-dependent, and convergence between methods is the safer indicator of robust features.